Experiment no.

Aim: Measurement of reactive power of an inductive load using single wattmeter method.

Instruments Required:

Sl.no	Name	Type	Range
1	Wattmeter	Dynamometer	5/10A, 200/400V
2	Ammeter	MI	0-10A
3	Voltmeter	MI	0-600V
4	3-Phase inductive load		
5	3-Phase Variac		0/400-400V 15A

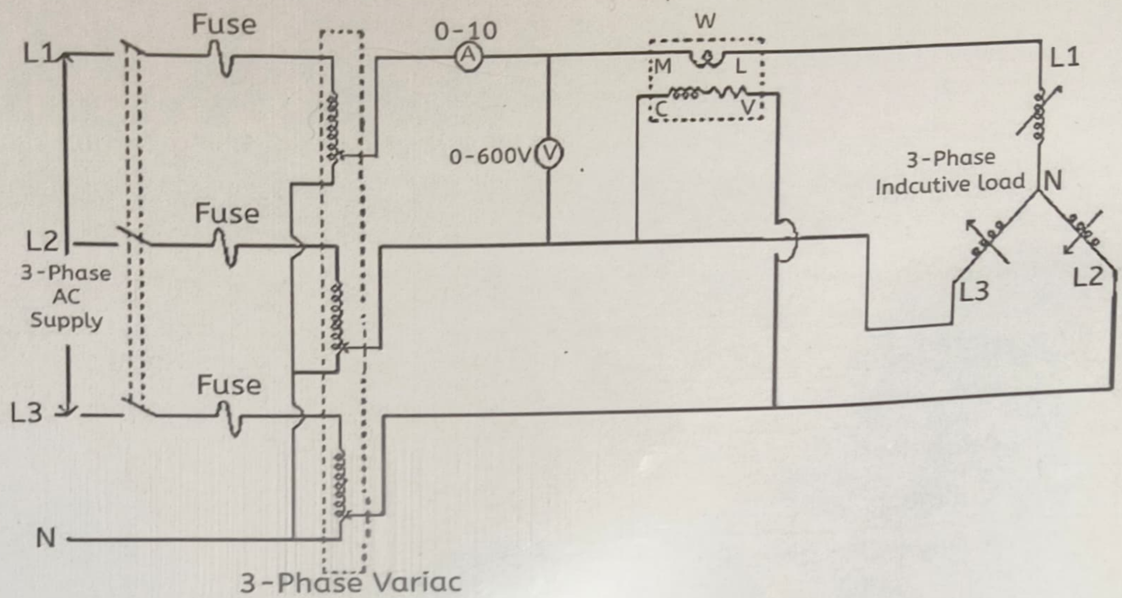
Theory: If the 3-phase load is balanced, then the reactive power drawn by the load can be determined by using a single wattmeter.

For this purpose, the current coil of the wattmeter is connected in one line and the potential coil is connected across the other two lines as shown in Fig. (a). It can be proved that the total reactive power drawn by the load is equal to $\sqrt{3}$ times the reading of the wattmeter.

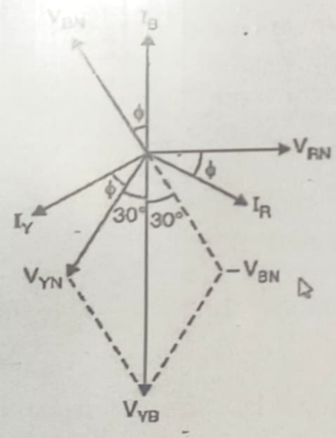
Fig. (a) shows the connections of a single wattmeter for measuring reactive power in a 3-phase balanced load having lagging p.f. cos Φ. Here current coil of the wattmeter is connected in the R-line and the potential coil is connected across the lines Y and B.

So, current through current coil = I_R

And P.D. across potential coil, $V_{YB} = V_{YN} - V_{BN}$ --------------- (phasor difference)



Circuit Diagram



Phasor Diagram

To obtain V_{YB}, find the phasor sum of V_{YN} and V_{BN} as shown in the phasor diagram in Fig. (b). It is clear from the phasor diagram that the phase angle between V_{YB} and I_R is (90°- Φ).

Therefore, Wattmeter reading, $W = V_{YB} I_R \cos(90° - \Phi) = V_{YB} I_R sin\Phi$

Since the load is balanced, $V_{YB} = V_L$, the line voltage and $I_R = I_L$, the line current.

Therefore, $$W = V_L I_L sin\Phi$$

Since, total reactive power (Q) of load is $Q = \sqrt{3} V_L I_L sin\Phi$

so $$Q = \sqrt{3} V_L I_L sin\Phi = \sqrt{3} W$$

or Reactive power = $\sqrt{3}$ x Wattmeter reading

Thus, if we multiply the wattmeter reading by $\sqrt{3}$, we get the total reactive power drawn by the 3-phase balanced load.

Procedure:

1. Connect the circuit as per the circuit diagram.
2. Keep the three phase variac at zero-volt position and inductive load at minimum position.
3. Switch on the three-phase power supply.
4. Now slowly vary the three phase variac to its rated voltage (400V) and note down the readings of the ammeter, voltmeter and wattmeter.
5. By increasing the inductive load and tabulates the readings of ammeter, voltmeter and Wattmeter up-to the rated current.
6. Now set the inductive load and three phase variac to its minimum position.
7. Switch off the three-phase supply.
8. Calculate the three-phase reactive power.

Observation Table:

Sl.no.	Current (in amp)	Voltage (in volt)	Wattmeter reading W	W X multiplying factor	Reactive power
1					
2					
3					
4					

Multiplying factor of Wattmeter=

Calculations: Reactive power (Q) = ?

Power factor cosΦ = ?

Result:

Precautions:

1. Live terminals should not be touched.
2. Connect the circuit without any loose connection.
3. Load current should not exceed rated current value.